

Meet the Parent Functions

Students will identify the 6 parent functions from their equations and graphs, and state each one's domain and range.

 Graded online lesson • 1.1: [Function Families](#)

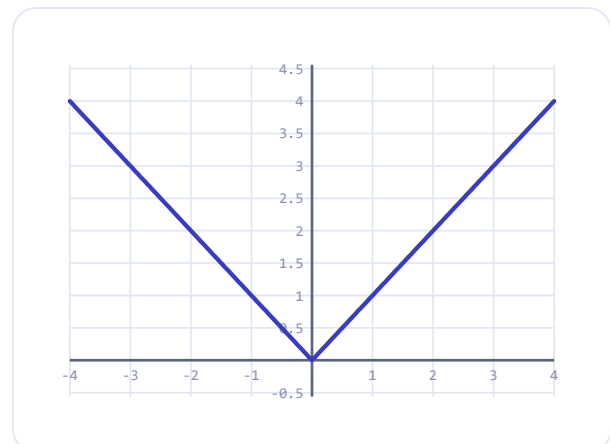
01 • PROJECT Launch deck — Meet the 6 Parent Functions

HOOK

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Name that graph

- ◆ A curve appears on screen — **no equation attached**.
- ◆ By the end of class you'll name its family, then state its domain and range on sight.
- ◆ Six shapes rule everything we do in Pre-Cal. Meet them today.



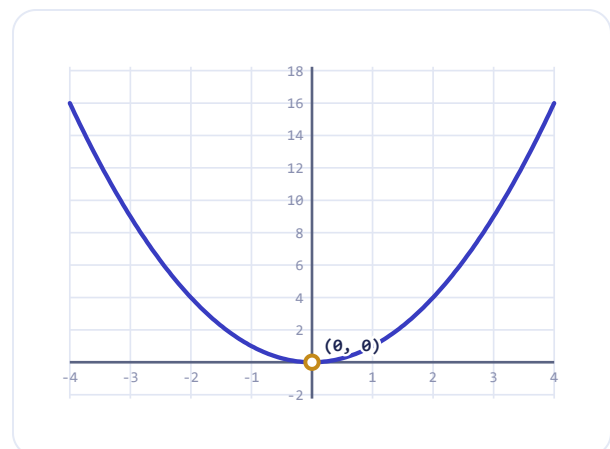
V shape or U shape? By 9:00 you'll never confuse them

NEW CONCEPT

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One family, one shape

- ◆ A **parent function** is the simplest member of a family — no shifts, no stretches.
- ◆ Every function in this course is a parent that's been stretched, shifted, or flipped.
- ◆ Recognize the parent → you already know the domain and range.



$y = x^2$ — the quadratic parent

Three words we use every day

Parent function — the simplest form of a family

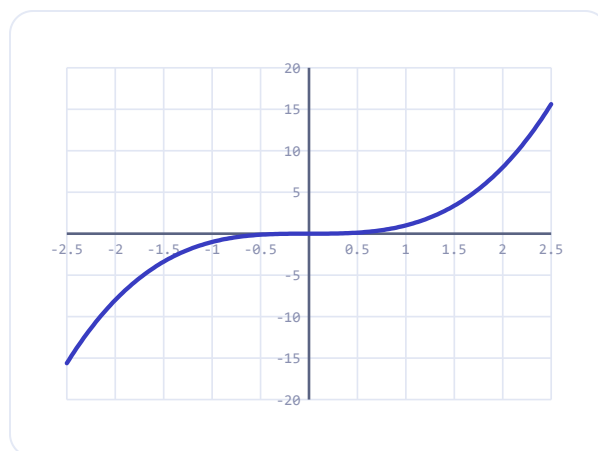
Domain — all allowed x-values (inputs)

Range — all possible y-values (outputs)

NEW CONCEPT

The 6 parents

- Linear: $f(x) = x$ · Quadratic: $f(x) = x^2$ · Cubic: $f(x) = x^3$
- Absolute value: $f(x) = |x|$ · Square root: $f(x) = \sqrt{x}$ · Reciprocal: $f(x) = \frac{1}{x}$
- Each has a signature shape AND a signature domain/range.



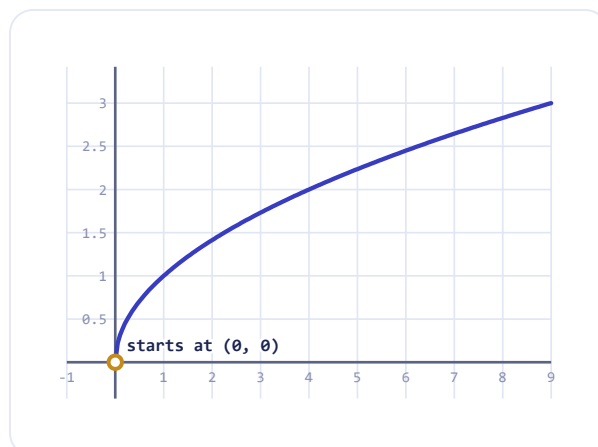
$y = x^3$ — S-curve through the origin

WORKED EXAMPLE

We do: $f(x) = \sqrt{x}$

WORKED EXAMPLE

- The radical symbol signals the **square root** parent.
- A radicand can't be negative: $x \geq 0$ — that's the domain.
- The output of a square root is never negative: $y \geq 0$ — that's the range.



$y = \sqrt{x}$ — starts at the origin, only goes right and up

Shifts don't change the family

COMMON MISTAKE

$f(x) = (x - 3)^2$ is still **quadratic** — the -3 only slides the graph right.

Added or subtracted numbers move a graph; they never change which family it belongs to.

KEEP IN MIND

Cover the shift with your hand and read the core operation: square? root? bars? fraction?

YOU TRY

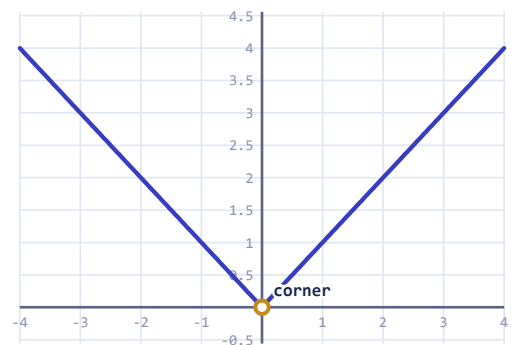
Your turn

YOU TRY

Name the family, domain, and range:

(a) $f(x) = x^3$

(b) $f(x) = |x|$



one of these two — which?

WHERE WE'RE HEADED

Where this week goes

- ◆ **Tue** — where a graph increases, decreases, and turns (max/min).
- ◆ **Wed** — domains that break: denominators and radicals.
- ◆ **Thu** — even, odd, or neither (symmetry by algebra).
- ◆ **Fri** — all four checks on one function + quiz-style practice.

Worked procedure — “I do”

1 Identify the parent function of $f(x) = x^2$ and give its domain and range.

1. The equation is x raised to the 2nd power, matching the quadratic parent.
2. There is no root and no denominator, so every real number is allowed: domain is all reals.
3. Since squaring never gives a negative result, the smallest output is 0.

ANSWER

Quadratic; domain: all real numbers; range: $y \geq 0$

2 Identify the parent function of $f(x) = |x|$ and give its domain and range.

1. Absolute value bars around x signal the absolute value parent.
2. Any real number can go inside absolute value bars, so the domain is all reals.
3. Absolute value output is never negative, so the range starts at 0.

ANSWER

Absolute value; domain: all real numbers; range: $y \geq 0$

3 Identify the parent function of $f(x) = \sqrt{x}$ and give its domain and range.

1. A radical (square root) symbol signals the square root parent.
2. The expression under the radical must be ≥ 0 , so $x \geq 0$.
3. The graph starts at $(0, 0)$ and only increases, so the range is $y \geq 0$.

ANSWER

Square root; domain: $x \geq 0$; range: $y \geq 0$

Guided notes — teacher key (students get blanks, printed 2-up)

The 6 Parent Functions

TEACHER KEY · Aug 10

Formula bank **Linear:** $f(x) = x$ **Quadratic:** $f(x) = x^2$ **Cubic:** $f(x) = x^3$ **Absolute value:** $f(x) = |x|$
Square root: $f(x) = \sqrt{x}$ **Reciprocal:** $f(x) = \frac{1}{x}$

VOCABULARY

Parent function — the **simplest** member of a family of functions --- no shifts, stretches, or reflections applied.

Domain — the set of all allowed **input** (x) values a function can use.

Range — the set of all **output** (y) values a function actually produces.

EXAMPLE 1 · TYPE 1 — IDENTIFY FAMILY, DOMAIN, RANGE

Identify the parent function of $f(x) = \frac{1}{x}$ and give its domain and range.

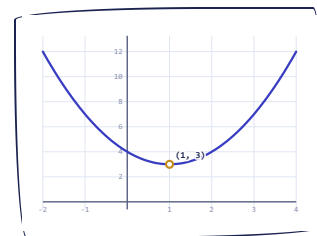
1. The rule is a fraction with x in the denominator, so this is the **reciprocal** parent function.
2. A denominator can never equal 0, so x cannot be 0: domain is $x \neq 0$.
3. Since $y = 1/x$ never produces an output of exactly 0 either, the range is $y \neq 0$.

EXAMPLE 2 · TYPE 2 — SHIFTED EQUATION, NAME THE PARENT

Name the **ORIGINAL** parent family of $g(x) = (x - 1)^2 + 3$.

1. The -1 and $+3$ only **shift the graph** — they never change the family.
2. The core operation is **squaring**, so the parent is the **quadratic** family: $f(x) = x^2$.

Quadratic — g is x^2 shifted right 1, up 3.



YOU TRY

1.

Identify the parent function of $f(x) = x^3$ and give its domain and range.

cubic; domain all real numbers, range all real numbers

2.

Identify the parent function of $f(x) = |x|$ and give its range.

absolute value; range $y \geq 0$

SUMMARY

To identify a parent function: match the **equation** to one of the 6 known shapes, then check for **restrictions** caused by a root or a denominator.

For each equation, name the parent function family, then state its domain and range using inequality notation.

TYPE 1 Identify the family, domain, and range**EXAMPLE**

$f(x) = \frac{1}{x}$ is the **reciprocal** parent. A denominator can't be 0, so the domain is $x \neq 0$; the graph never touches $y = 0$ either, so the range is $y \neq 0$.

Directions: Name the parent family, then give the domain and range.

1. $f(x) = x$
2. $f(x) = x^2$
3. $f(x) = x^3$
4. $f(x) = |x|$
5. $f(x) = \sqrt{x}$
6. $f(x) = \frac{1}{x}$

TYPE 2 Match a shifted equation back to its parent family**EXAMPLE**

$g(x) = \sqrt{x} + 4$ still has a radical, so its parent family is **square root** (the +4 just shifts it up 4).

Directions: Even though these are shifted, name the **ORIGINAL** parent family only.

7. $g(x) = (x - 2)^2$
8. $g(x) = |x + 1| - 3$
9. $g(x) = \frac{1}{x - 5}$
10. $g(x) = (x + 3)^3 + 1$

CHALLENGE Stretch problems (optional)

11. Sketch all 6 parent functions from memory on one set of axes, labeling each with its equation.

IDENTIFY THE FAMILY, DOMAIN, AND RANGE

1. Linear; domain: all real numbers; range: all real numbers
2. Quadratic; domain: all real numbers; range: $y \geq 0$
3. Cubic; domain: all real numbers; range: all real numbers
4. Absolute value; domain: all real numbers; range: $y \geq 0$
5. Square root; domain: $x \geq 0$; range: $y \geq 0$
6. Reciprocal; domain: $x \neq 0$; range: $y \neq 0$

MATCH A SHIFTED EQUATION BACK TO ITS PARENT FAMILY

7. Quadratic (the $(x - 2)^2$ is still squared, just shifted right)
8. Absolute value (the bars are still there, just shifted)
9. Reciprocal (still a fraction with x in the denominator)
10. Cubic --- the exponent of 3 on $(x + 3)$ shows this came from the cubic parent, just shifted left 3 and up 1

CHALLENGE

11. Six correct sketches on one grid: $y = x$ (straight line through origin, slope 1); $y = x^2$ (U-shaped parabola, vertex $(0, 0)$); $y = x^3$ (S-shaped curve through origin, rising left to right); $y = |x|$ (V-shape, vertex $(0, 0)$, opening up); $y = \sqrt{x}$ (half-parabola starting at $(0, 0)$, in the first quadrant only); $y = \frac{1}{x}$ (two hyperbola branches in quadrants I and III, never touching either axis). Each curve labeled with its equation; $\sqrt{x}$ and $1/x$ should visibly show their restricted domains.

1. Name the parent family of $g(x) = (x - 2)^3$, then state the domain and range of that PARENT function.